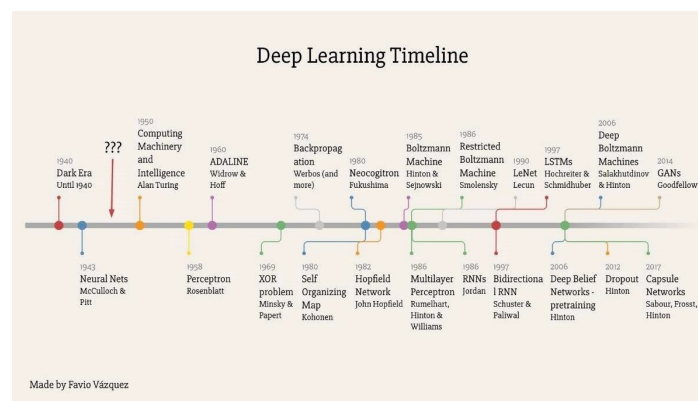


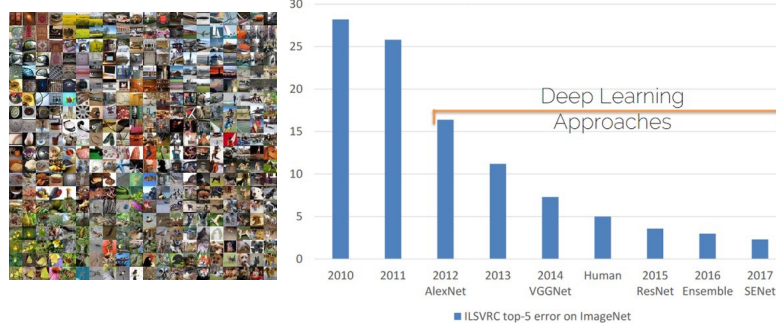
Deep learning software

Deep learning is old



The deep learning revolution

🔥 In 2012 the Alexnet paper crushed all competition on in the imagenet benchmark



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Since then

Q: What is your favorite animal?
A: My favorite animal is a dog.
Q: Why?
A: Because dogs are loyal and friendly.
Q: What are two reasons that a dog might be in a bad mood?
A: Two reasons that a dog might be in a bad mood are if it is hungry or if it is hot.
Q: How many eyes does a giraffe have?
A: A giraffe has two eyes.
Q: How many legs does a frog have?
A: A frog has four legs.
Q: Are there any animals with three legs?
A: No, there are no animals with three legs.
Q: Why don't animals have three legs?
A: Animals don't have three legs because they would fall over.

🔥 ChatGPT:
Generation of text

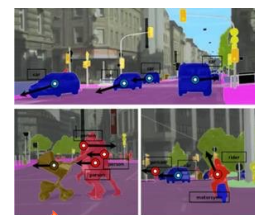


🔥 Stable diffusion:

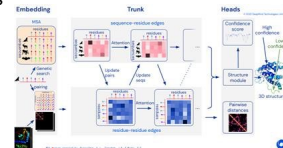
Text to image translation/generation



🔥 AlphaGO:
Beating humans at their own games



🔥 U-net:
Real-time semantic segmentation



🔥 Alphafold:
Solving protein engineering

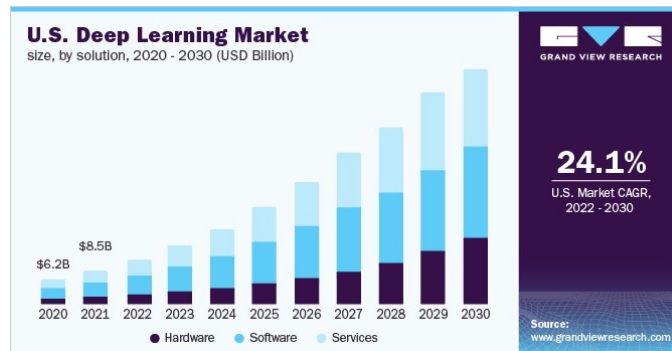


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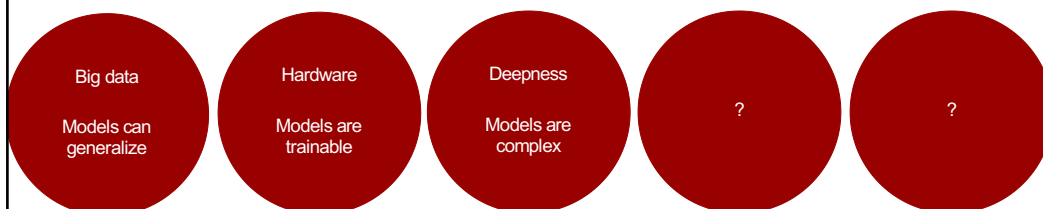
Why you should care about deep learning

💡 Usage of Deep Learning in industry is increasing very fast!



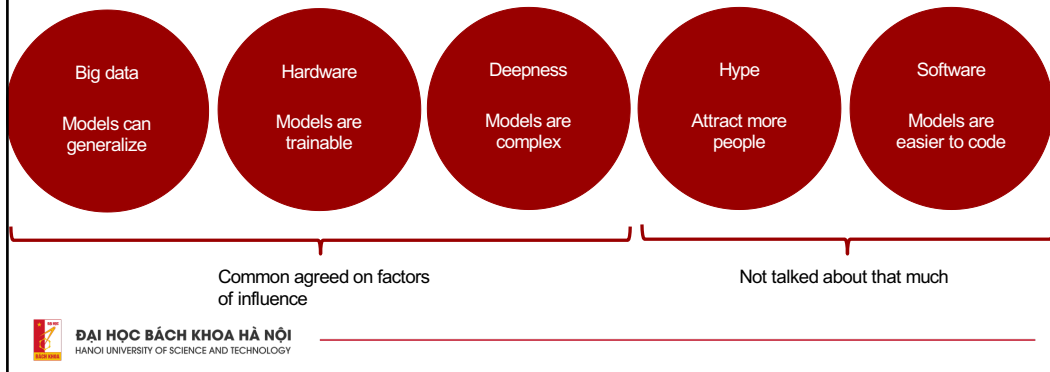
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The drivers of the revolution



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The drivers of the revolution



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Why do we need specialized frameworks for DL?

Deep learning is just a log of simple math

- 💡 But we need to do it efficiently
- 💡 We need to take care of hardware acceleration (e.g. it can run on GPU)
- 💡 We need to take care of gradient backpropagation
- 💡 Optimizers, data interface etc. also complicates implementation

We do not want to deal with this **cursed** 🧛

```
import numpy as np

class Linear(object):
    def __init__(self, input_dim: int, num_hidden: int = 1):
        self.weight = np.random.randn(input_dim, num_hidden)
        self.bias = np.zeros(num_hidden)

    def __call__(self, x):
        self.x = x
        output = x @ self.weight + self.bias
        return output

    def backward(self, gradient):
        self.weight_gradient = self.x.T @ gradient
        self.bias_gradient = gradient.sum(axis=0)
        self.x_gradient = gradient @ self.weight
        return self.x_gradient

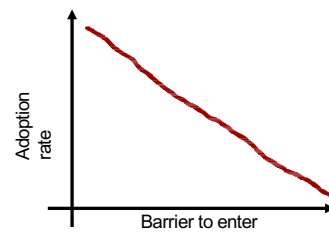
    def update(self, lr):
        self.weight = self.weight - lr * self.weight_gradient
        self.bias = self.bias - lr * self.bias_gradient

if __name__ == "__main__":
    x = np.random.randn(10, 5)
    layer = Linear(5, 1)
    y = layer(x)
    grad = layer.backward(np.ones((10, 5)))
    layer.update(1e-2)
```

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Barrier to enter

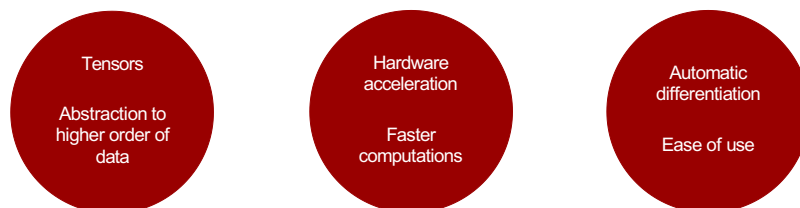
- 💡 Without proper DL frameworks, ML/DL/AI as a field would have a very high barrier to enter
- 💡 Low barrier to enter mean it accessible to more people = more people driving the technology forward
- 💡 AI would be gated from the public (trustworthy AI)



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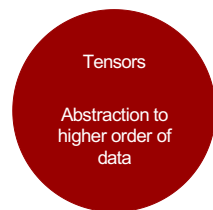
How to make a modern DL framework



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How to make a modern DL framework



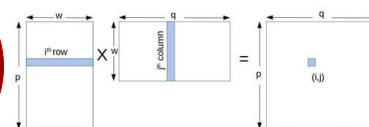
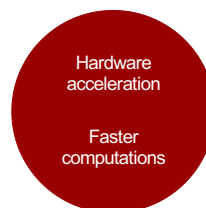
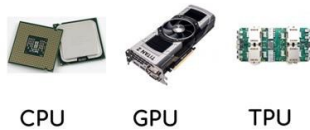
A tensor is an N-dimensional array of data



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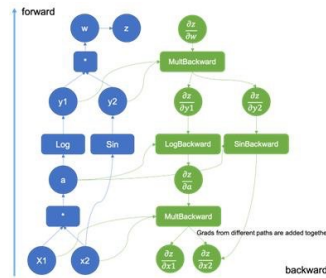
How to make a modern DL framework



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How to make a modern DL framework



Automatic
differentiation

Ease of use

```
(base) C:\Users\msde>python
Python 3.8.5 (default, Sep 3 2020, 21:29:08) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> import torch
>>> z=torch.ones(5, requires_grad=True)
>>> z.backward(torch.ones(5, requires_grad=True), grad_fn=<MulBackward0>)
>>>
```

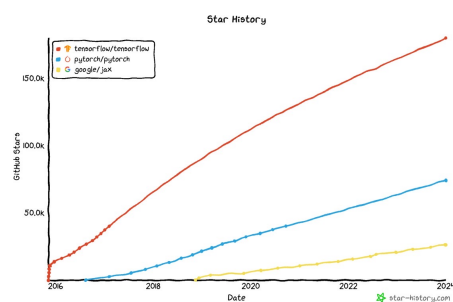
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The current landscape

Pytorch vs Tensorflow vs Jax all support the same underlying feature set

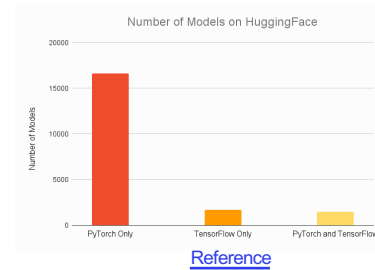
- ⚡ Easy to use Python interface
- ⚡ Hardware acceleration
- ⚡ Research and industry specific features



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The current landscape

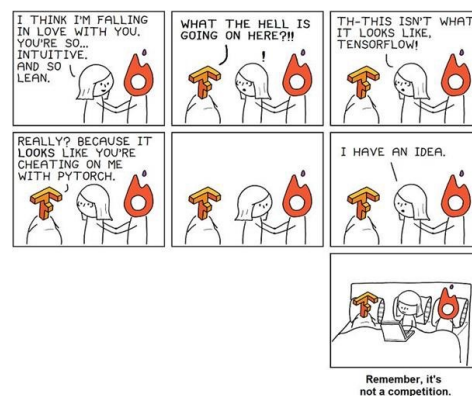


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I highly recommend...

If you have the time, learn the basics of them all 😊



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In practice, people often use high-level frameworks

Makes a lot of coding much easier.

Recommend, to only use these if you understand the underlying framework.

We get back top one of these.



Haiku



Sonnet

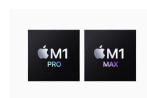


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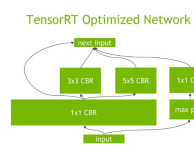
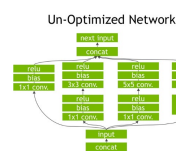
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Current trends in deep learning software

⚡ More accelerators



⚡ Compiled models



⚡ Lower precision computations



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